

Title

MySQL Database Backup and Recovery

Objectives

1. To understand the importance of database backup.
 2. To learn different types of MySQL backups.
 3. To perform database backup using command-line tools.
 4. To restore a MySQL database from backup.
 5. To ensure data security and disaster recovery.
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Problem Statement

Design and implement a backup and recovery strategy for a MySQL database to ensure:

- Data protection against accidental deletion.
 - Recovery from system failure.
 - Business continuity through reliable restore procedures.
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Outcomes

After completing this experiment, students will be able to:

- Perform logical backups using MySQL tools.
 - Restore databases from backup files.
 - Understand full and incremental backups.
 - Handle disaster recovery scenarios.
 - Verify data integrity after restoration.
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Software and Hardware Requirements

Software

- MySQL
- MySQL Workbench / Command Prompt / Terminal

Hardware

- Minimum 4GB RAM
- 10GB free disk space
- Windows/Linux/Mac OS

Theory Concept (Brief)

◆ Backup

A backup is a copy of database data stored separately to prevent data loss.

◆ Recovery

Recovery is the process of restoring data from a backup after failure.

◆ Types of Backups in MySQL

1. **Logical Backup**
 - Uses mysqldump
 - Produces SQL file
 - Portable and easy to restore
 2. **Physical Backup**
 - Copies actual data files
 - Faster for large databases
 3. **Full Backup**
 - Entire database backup
 4. **Incremental Backup**
 - Only changed data since last backup
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Sample Database

Database Name: salesdb

Table: customers

customer_id name city purchase_amount

Implementation

1 Create Sample Database

```
CREATE DATABASE salesdb;  
USE salesdb;
```

```
CREATE TABLE customers (
```

```
customer_id INT PRIMARY KEY,  
name VARCHAR(50),  
city VARCHAR(50),  
purchase_amount DECIMAL(10,2)  
);
```

```
INSERT INTO customers VALUES  
(1,'Rahul','Mumbai',5000),  
(2,'Sneha','Delhi',7000),  
(3,'Amit','Pune',6000);
```

2 Take Full Backup Using mysqldump

◆ Backup Entire Database

```
mysqldump -u root -p salesdb > salesdb_backup.sql
```

This creates a file:
salesdb_backup.sql

3 Verify Backup File

Open the file to confirm it contains:

- CREATE DATABASE statements
 - CREATE TABLE statements
 - INSERT statements
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4 Simulate Data Loss

```
DROP DATABASE salesdb;
```

5 Restore Database from Backup

```
mysql -u root -p < salesdb_backup.sql
```

6 Verify Recovery

```
USE salesdb;  
SELECT * FROM customers;
```

Output Table After Recovery

customer_id	name	city	purchase_amount
1	Rahul	Mumbai	5000
2	Sneha	Delhi	7000
3	Amit	Pune	6000

Algorithm

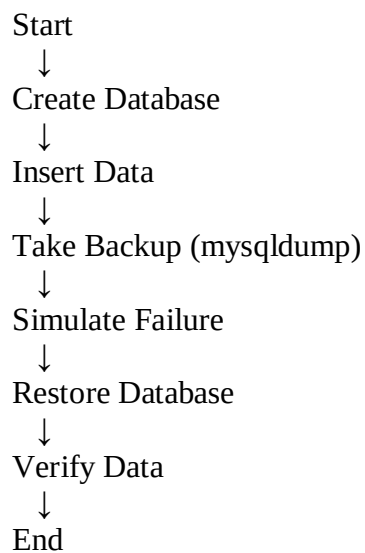
Backup Process

1. Start
2. Identify database
3. Use mysqldump command
4. Store backup file securely
5. End

Recovery Process

1. Start
 2. Identify backup file
 3. Use mysql restore command
 4. Verify restored data
 5. End
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Flowchart



Test Cases

Test Case	Operation	Expected Result	Status
TC1	Take backup	Backup file created	Pass
TC2	Drop database	Database removed	Pass

TC3	Restore database	Database restored	Pass
TC4	Verify records	All records available	Pass

Test Data Set

customer_id	name	city	purchase_amount
1	Rahul	Mumbai	5000
2	Sneha	Delhi	7000
3	Amit	Pune	6000

Advantages of Backup

- Prevents data loss
 - Enables disaster recovery
 - Ensures business continuity
 - Protects against hardware failure
 - Helps recover from accidental deletion
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Conclusion

The MySQL database backup and recovery process was successfully implemented. Using my sql dump, a full logical backup was created and restored after simulating data loss. This experiment highlights the importance of regular backups in ensuring data security and disaster recovery. Proper backup strategies help maintain business continuity and system reliability.
